

Content Determination of Ammonium Chloride

Method for the alkalimetric content determination of Ammonium Chloride

Ammonium Chloride is an expectorant. It promotes secretion production in the bronchi and accelerates the release of the secretions. It also promotes the elimination of lead and basic drugs (amphetamines) and is therefore used in cases of poisoning.

Sample	Ammonium Chloride, NH ₄ Cl	Structure	
Substance	Ammonium Chloride M = 53.49 g/mol	Literature	Ph. Eur. 1997; Monograph "Ammonium Chloride"
Chemicals	Formaldehyde solution 35% R Deionised water	Sample preparation Add 40 mL formaldehyde solution into a titration beaker and neutralise with 0.1M NaOH. Weigh 0.99 to 1.01 g ammonium chloride in a 100 mL titration beaker. Add 40 mL deionised water and 5 mL of the neutralised formaldehyde solution.	
Titrant	c(NaOH) = 1 mol/L c(NaOH) = 0.1 mol/L		
Standard	Potassium hydrogen phthalate (M521)		
Instruments	METTLER TOLEDO DL70ES/DL77 METTLER TOLEDO DL58/55/53/50Graphix		
Accessories	Titration beaker ME-101974 SP250 Peristaltic pump Rondo60 Sample changer		
Indication	DG111-SC	Remarks The previously determined loss on drying content is saved as the auxiliary value H2O.	
Chemistry	1. Formaldehyde reacts with ammonium chloride to give hexamethylenetetramine while releasing hydrochloric acid: $4\text{NH}_4\text{Cl} + 6\text{H}_2\text{CO} \rightarrow (\text{CH}_2)_6\text{N}_4 + 4\text{HCl} + 6\text{H}_2\text{O}$ 2. Titration of hydrochloric acid with NaOH: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$		
Calculation	$\text{R}(\%) = (\text{Q} \cdot \text{M} / (10 \cdot \text{m})) + \text{H2O}$ Q = Titrant consumption in mmol M = Molar mass Ammonium Chloride m = Sample size in g H2O = % Loss on drying of Ammonium Chloride	Waste disposal	Halogen-free organic solvents waste.
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Results

METTLER DL77

Titration V3.1

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Method: G053 Cont. Ammonium chloride Date: 08-09-1999 11:42
User: **Pludra** Measured: 08-09-1999 13:57

RESULTS

No.	ID1	Sample size and results			
1/1	992224	1.0072 g	R1 = 99.70	%	NH4Cl
1/2	992224	0.9923 g	R1 = 99.46	%	NH4Cl
1/3	992224	1.001 g	R1 = 99.45	%	NH4Cl
1/4	992224	1.0024 g	R1 = 99.41	%	NH4Cl
1/5	992224	0.9921 g	R1 = 99.41	%	NH4Cl
1/6	992224	0.9932 g	R1 = 99.38	%	NH4Cl

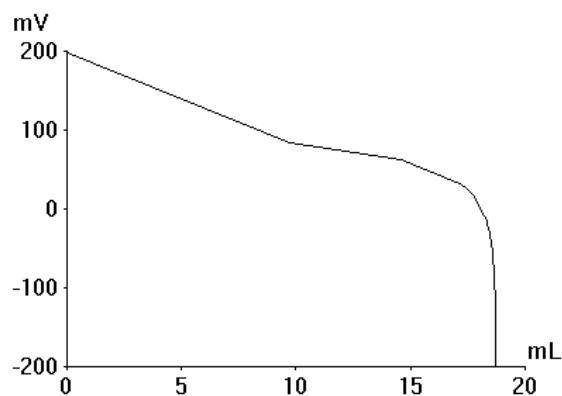
STATISTICS

Number of results R1 n = 6
Mean value $\bar{x}$ = 99.47 % NH4Cl
Standard deviation s = 0.119971 % NH4Cl
Rel.standard deviation srel = 0.121 %

Table of measured values

EQP	ET	Volume mL	Signal mV	1st Deriv. mV/mL	Time min:s
EQP1	ET1	0.0000	198.0		0:17
		9.7140	83.7	-11.8	0:57
		14.5710	61.5	-4.6	1:53
	ET2	17.0000	34.0	-11.3	2:18
		17.2000	30.7	-16.8	2:29
		17.4000	25.8	-24.5	2:37
		17.5550	22.5	-21.2	2:49
		17.7550	15.1	-36.7	2:54
		17.8500	11.5	-38.3	2:58
		17.9750	5.5	-47.6	3:03
		18.0620	1.4	-47.5	3:09
		18.1670	-4.8	-58.7	3:13
		18.2380	-9.2	-63.1	3:19
		18.3120	-15.1	-79.5	3:23
		18.3640	-19.6	-86.2	3:30
		18.4170	-24.8	-97.7	3:36
		18.4620	-29.9	-113.6	3:42
		18.5000	-35.8	-154.7	3:46
		18.5250	-40.2	-179.2	3:53
		18.5490	-44.7	-183.7	3:59
		18.5750	-51.4	-261.2	4:08
		18.5950	-57.3	-294.0	4:16
		18.6150	-64.8	-374.5	4:25
		18.6350	-75.2	-521.5	4:35
		18.6550	-90.4	-756.0	4:48
		18.6750	-112.9	-1127.0	5:01
		18.6950	-134.5	-1081.5	5:14
		18.7150	-151.3	-840.0	5:25

Titration curve



Method

Method: G053 Cont. Ammonium chloride
Version: 08-09-1999 11:42

Title

Method ID G053
Title Cont. Ammonium

Chloride

Date/time 08-Sep-1999 11:02

Sample

Number samples 3
Titration stand ST20 1
Entry type Weight m
Lower limit [g] 0.99
Upper limit [g] 1.01
ID1
Molar mass M 53.49
Equivalent number z 1
Temperature sensor Manual

Stir

Speed [%] 50
Time [s] 10

Titration

Titrant NaOH
Concentration [mol/L] 1.0
Sensor DG111-SC
Unit of meas. mV
Titration mode EQP
Predispensing 1 mL
Volume [mL] 17.0
Titrant addition DYN
dE(set) [mV] 5.0
Limits dV Absolute
dV(min) [mL] 0.02
dV(max) [mL] 0.2
Measure mode EQU
dE [mV] 0.25
dt [s] 1.0
t(min) [s] 2.0
t(max) [s] 20.0
Threshold 500.0
Maximum volume [mL] 20.0
Termination after n EQPs Yes
n = 1
Evaluation procedure Standard
Steepest jump only Yes

Calculation

Result name NH4Cl
Formula $R = (Q \cdot C / m) + H_2O$
Constant $C = M / (10 \cdot z)$
Result unit %
Decimal places 3

Rinse

Auxiliary reagent Water
Volume [mL] 20.0

Record

Output unit Printer+Computer
Raw results last sample Yes
Table of values Yes
E - V curve Yes

Statistics

Ri (i=index) R1
Standard deviation s Yes
Rel. standard deviation srel Yes

Record

Output unit Printer+Computer
All results Yes